

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Botany

Paper: BOB-601 — Plant Metabolism, Biochemistry and Biotechnology

Time: Three hours

Maximum Marks: 70

Note: Attempt Five questions in all. The marks of each question have been indicated at the right hand margin. Question No.1 is compulsory.

Question 1. Fill in the blanks and answer (note more than 50 words each) the following:
[1 × 10 = 10]

- (a) Ri plasmid is found in _____
- (b) _____ is the site of starch synthesis in leaves.
- (c) What is EF-Tu?
- (d) Define rut site.
- (e) Cellulose is a linear polymer of _____ linked _____ units.
- (f) Define rho factor.
- (g) Define T_m.
- (h) What is Poly (A) polymerase?
- (i) Calculate the T_m of given DNA sequence: ATGCTTAATTGCTA
- (j) What is the function of telomerase?

Question 2. Write in detail about the mechanism of mRNA processing in eukaryotes. [15]

Question 3. Discuss various types of post-translational modifications. Describe the secondary structure of the protein. [15]

Question 4. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Components and role of nitrogenase enzyme complex
- (b) Enzymes involved in the metabolism of nitrate into ammonia
- (c) Initiation of translation in prokaryotes

Question 5. Describe the detailed mechanism of *Agrobacterium*-mediated gene transfer for the production of transgenic plants. [15]

Question 6. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Southern blotting
- (b) SDS-PAGE
- (c) Cloning of a gene

Question 7. Briefly describe any three of the following:

[5 × 3 = 15]

- (a) Name and function of different proteins involved in DNA replication
- (b) Different forms of DNA and their characteristics
- (c) Various models to explain the mode of action of enzymes
- (d) Photoreactivation

Question 8. Write short notes on any two of the following:

[7.5 × 2 = 15]

- (a) ATP synthesis
- (b) Isoelectric focusing
- (c) Important features of Watson and Crick DNA model

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